UL Benchmarks technical test

Goal: Create a RAG pipeline using Phi-3.5-mini-instruct model on any platform of your choice, accelerate the inference on two different accelerators, and analyze the results.

Introduction

Retrieval-Augmented Generation (RAG) is the process of optimizing the output of a large language model (LLM) to enable it to use an external knowledge base as reference to generate its response. A common use case of RAG in organizations is to enable LLMs to incorporate their internal knowledge base (i.e. documents) so that it can provide answers relevant to the organization without requiring the model to be retrained with the organization's data.

In this task, you are required to create a command line interface (CLI) app that runs Phi-3.5-mini-instruct model locally/on-device and enables a user to ask questions based on an external knowledge base with the use of the RAG framework.

Instructions

- Take the Phi-3.5-mini-instruct model from https://huggingface.co/microsoft/Phi-3.5-mini-instruct

- Create a demo CLI app that enables us to query the model using prompts relevant to the document shared.

- The programming language can be either C++ or Python.

Evaluation

- Provide a zip of the source code and documentation. You may also upload it to Github account and share a link.

- The repository must include instructions on how to compile and run the demo app. You do not need to provide the models if you have well documented steps on how we could reproduce them.

Bonus

- Collect relevant metrics of the LLM in the output (model load, ttft, etc.) to compare and analyze the performance of the two accelerators.